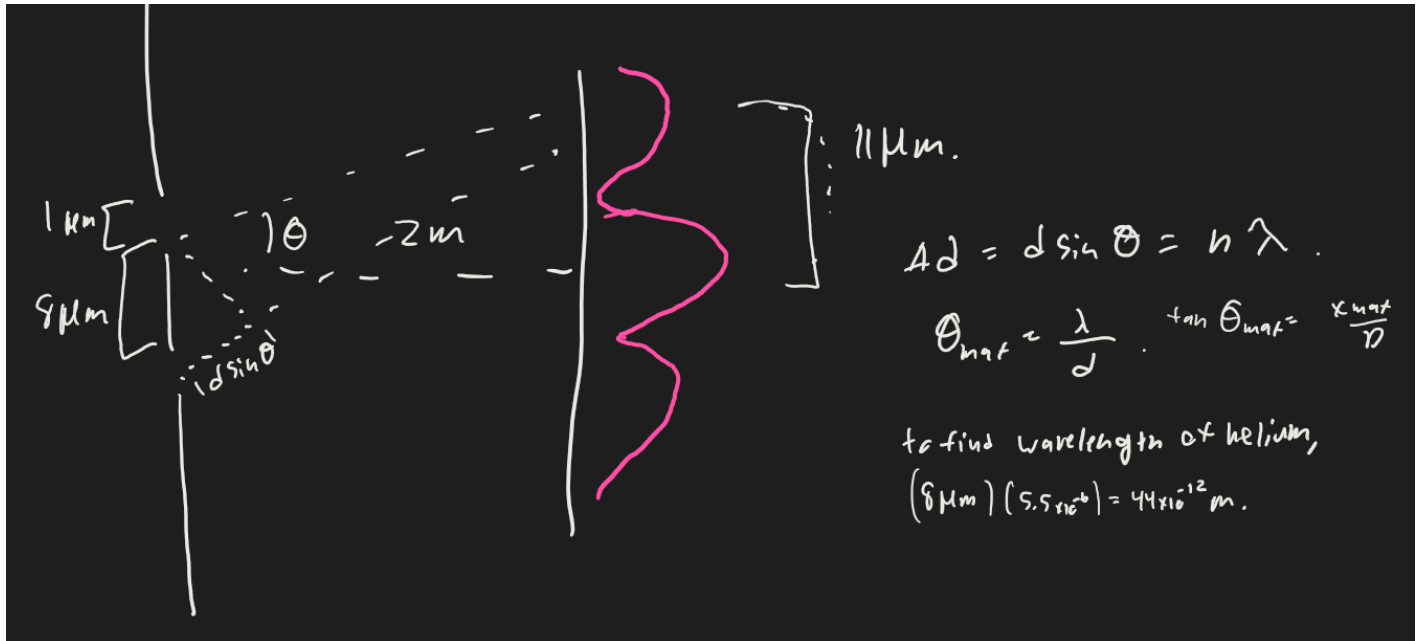


Atom Interferometry

Last class :(

Electrons and helium atoms can diffract and interfere. All matter has a De Broglie wavelength, $\lambda_{db} = \frac{h}{p}$.

Lets think about the interference patterns of helium.



For photons, we have (given planks constant h and the momentum of the photon p). $E = 6.67 * 10^{-34}$.

$$E = h\nu = \frac{hc}{\lambda}$$

$$E = pc,$$

$$p = \frac{h}{\lambda},$$

$$\lambda = \frac{h}{p}$$

We can use a similar result to calculate for the helium atom.

$$v_{he} = \frac{2.2 km}{s}, m_{he} = 4 * 1.67 * 10^{-27}$$

$$\lambda_{he} = \frac{h}{(mv)} = 45 \text{ pico meters. This matches up with the experimental result!}$$

The smaller wavelength gives us higher measurement resolution.

Probability of a photon diffracting is given by amplitude z ,

$$Prob = \sum_{paths} z_i z_i^*$$

For electrons,

$$Prob = *$$

$$i\hbar \frac{\partial(r,t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2(r,t) + v(r)(r,t)$$